

Geophysics 3A — Week 8 Workshop

Geothermics: Steady-State Heat Flow

May 5, 2026

Workshop Objectives

By the end of this workshop, students should be able to:

- Use conservation of energy to reason about steady-state geotherms.
- Explain how thermal conductivity contrasts redistribute isotherms.
- Interpret geotherm curvature in terms of crustal heat production.
- Relate conductivity and heat production variations to geothermal potential.

0:00–0:15 Conceptual Framing

- Steady-state heat flow as an energy bookkeeping problem.
- Review Fourier's law and the meaning of heat flow.
- Emphasize the separation of roles:
 - conductivity controls *gradients*,
 - heat production controls *curvature*.

0:15–0:45 Refraction Exercise I: Insulating Sediments

Setup Compare two crustal columns with identical basal heat flow:

- Case A: crystalline crust exposed at the surface.
- Case B: crystalline crust overlain by a low-conductivity sedimentary layer.

Activities

- Predict temperature gradients in sediments vs basement.
- Sketch qualitative isotherms for each case.
- Identify where isotherms compress and where they stretch.

Key Concepts

- Heat flow conserved across layers.
- Lower conductivity → steeper gradient.
- Sediments act as a thermal blanket.

Geological Discussion

- Implications for geothermal gradients measured in boreholes.
- Relative geothermal potential of sedimentary basins vs exposed cratons.

0:45–1:20 Refraction Exercise II: Layered Crustal Materials

Focus Thermal refraction arising from conductivity contrasts within the crust.

Examples

- Shales (low k) vs quartzites (high k).
- Cyclic sedimentary layering preserved through metamorphism.

Activity

- Apply $q = -k\nabla T$ across layered media.
- Examine how gradients vary layer by layer.
- Discuss effective anisotropy in cyclically layered strata.

Connection

- Tie back to Week 2 physical-properties mixing demos.
- Discuss refraction in tilted and folded strata.

1:20–1:30 Break

1:30–2:05 Energy Conservation and Geotherm Curvature

Central Question How can two terranes with identical surface heat flow have different Moho temperatures?

Steps

- Write the energy balance:

$$q_0 = q_b + \int_0^{z_c} A(z) dz$$

- Infer relative crustal heat production given different basal inputs.
- Introduce the steady-state solution and the curvature term:

$$T(z) = T_0 + \frac{q_0}{k}z - \frac{A}{2k}z^2$$

Key Insight Curvature differences imply different deep temperatures even when surface gradients match.

2:05–2:40 Why Is Archean Heat Production So Low?

Observation Decay alone predicts high early heat production, yet preserved Archean crust shows limits.

Guided Discussion Topics

- Differentiation and partitioning of LILEs (K, Th, U).
- Selective erosion of felsic upper crust.
- Reworking: remelting and redistribution of older crust.
- Foundering of dense mafic roots.
- Secular mantle thermal evolution.

2:40–3:00 Synthesis

- Summarize roles of k and A .
- Connect observations to upcoming steady-state problem set.
- Preview transition to transient heat flow.